PRODUCT REQUIREMENTS DOCUMENT

Smart Cart *Companion*

An AI grocery planner that helps Walmart+ households plan the week's groceries within a real budget.

VARUN SAGAR K · AI GROCERY PLANNING FOR WALMART+ · BUDGET-CART-COMPANION.VERCEL.APP

+25%	+10–15%	+8–12%	<3s
NORTH STAR TARGET	BASKET CONVERSION	WALMART+ RETENTION	BASKET GENERATION

TL;DR

Grocery shoppers plan across separate tools: weekly flyers, a budgeting app, and their own hunt for savings. The result is overspending, missed flyer savings and decision fatigue. Walmart is excellent at fulfilling orders, but it doesn't help households *plan*. **Smart Cart Companion** is an AI grocery planner inside the Walmart experience. It turns this week's flyer and rollback deals into a personalized basket, keeps that basket within the household's real budget (typed in, or pulled from YNAB), and hands it off to the Walmart cart. Groceries are bought every week, so the product is measured on habit: the North Star is **Weekly Repeat Grocery Planning Usage**, with a target of +25% in the first six months of a Walmart+ beta. The business payoff is Walmart+ retention. The aim is to move Walmart from a place where groceries are bought to the place where households plan them.

SECTION 1

Problem & User

Product. Smart Cart Companion: a Walmart+ capability that plans a household's groceries within its real budget, before it shops.

Who it's for

- **Budget-conscious households** that need the weekly shop to fit a fixed amount
- **Students** on tight, predictable budgets
- **Meal preppers** who plan a week of food at once
- **Walmart+ subscribers**, the group whose retention the product is designed to lift

These segments come from competitor analysis, observed grocery-planning behavior, product gap analysis and market opportunity assessment. They still need validation with real users during the beta.

Today's workflow

STEP	WHAT SHOPPERS DO TODAY	RESULT
1	Browse the weekly flyers to find this week's deals	**Missed flyer savings**
2	Check the grocery budget in a separate budgeting app	**Overspending**
3	Work out by hand which items give the best value across both	**Decision fatigue**

Walmart's product gap

Walmart excels at transaction fulfillment but lacks four things a planning household needs: **proactive flyer intelligence**, **budget-aware planning**, **household optimization** and **a link to the household's financial tools**.

Why existing options fall short

OPTION	STRONG AT	WHY IT FALLS SHORT HERE
Walmart (native)	Commerce infrastructure, broad product range	No proactive budget intelligence, flyer automation or household grocery planning
Flipp	Flyer aggregation and finding savings	Doesn't build a personalized basket or complete the purchase
Instacart	Convenient delivery and shopping help	Little budget-first planning, weak financial integration
Amazon Fresh	Commerce and logistics	No flyer optimization or household budget alignment
YNAB	Budgeting	Doesn't turn the budget into an actual grocery plan
Smart Cart Companion	Combines grocery planning, flyer automation, budget sync, household personalization and repeat-purchase intelligence in one flow	

The opportunity

Make Walmart+ the default place where households plan their groceries. Solving the weekly budget, savings and repeat-shopping problems up front lets Walmart expand from commerce into household grocery and financial planning.

SECTION 2

Goals, Scope & User Flows

Vision. Plan groceries smarter within budget.

Long-term goal. A household-first AI planning ecosystem that automates flyer savings, aligns with real budgets, encourages healthier decisions, builds repeat weekly grocery habits and strengthens Walmart+ retention.

Goals

FOR	GOAL
Walmart	Lift Walmart+ retention by 8–12% among AI Planner users in the first six months · Lift AI basket-to-cart conversion by 10–15% · Lift flyer savings utilization by 25%
Shoppers	Stay within the grocery budget · Save planning time · Reduce overspending · See savings clearly · Make repeat grocery shopping simple

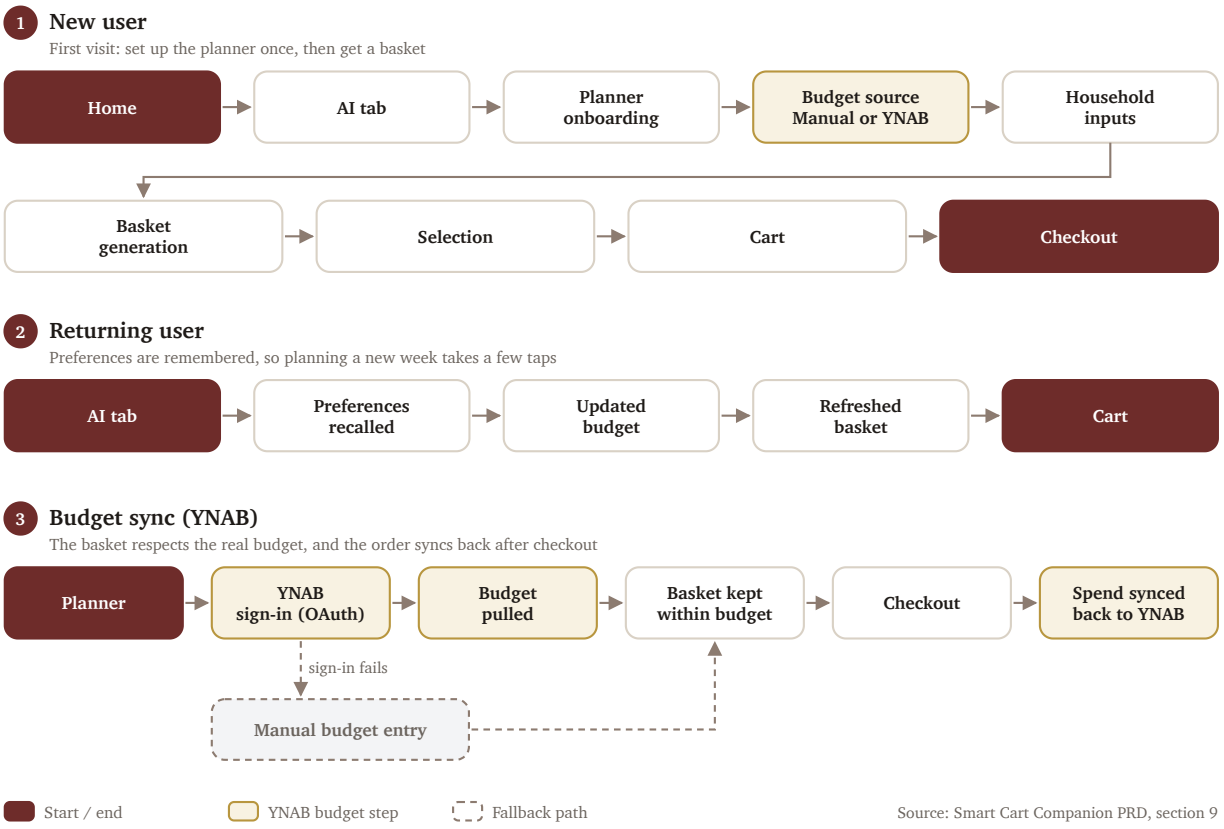
Scope

IN THE MVP	OPTIONAL POWER FEATURES	CONCEPTUAL (NOT BUILT)
AI Grocery Planner Personalized onboarding Household-size intelligence Dietary optimization Flyer/rollback ingestion (mock data, scalable design) YNAB budget sync Basket generation Budget validation Walmart-native cart handoff	KOHO balance awareness Smart Repeat Tiny Wins Pantry intelligence Advanced wellness Full predictive grocery automation	Walmart live product APIs Walmart flyer APIs Full KOHO / Plaid integration Live payments and order processing

Prioritization (RICE)

TIER	FEATURES
Highest	AI Planner · Budget Sync · Flyer Automation
Medium	Smart Repeat · Wellness optimization
Lower	Behavioral insights · Pantry intelligence

User flows



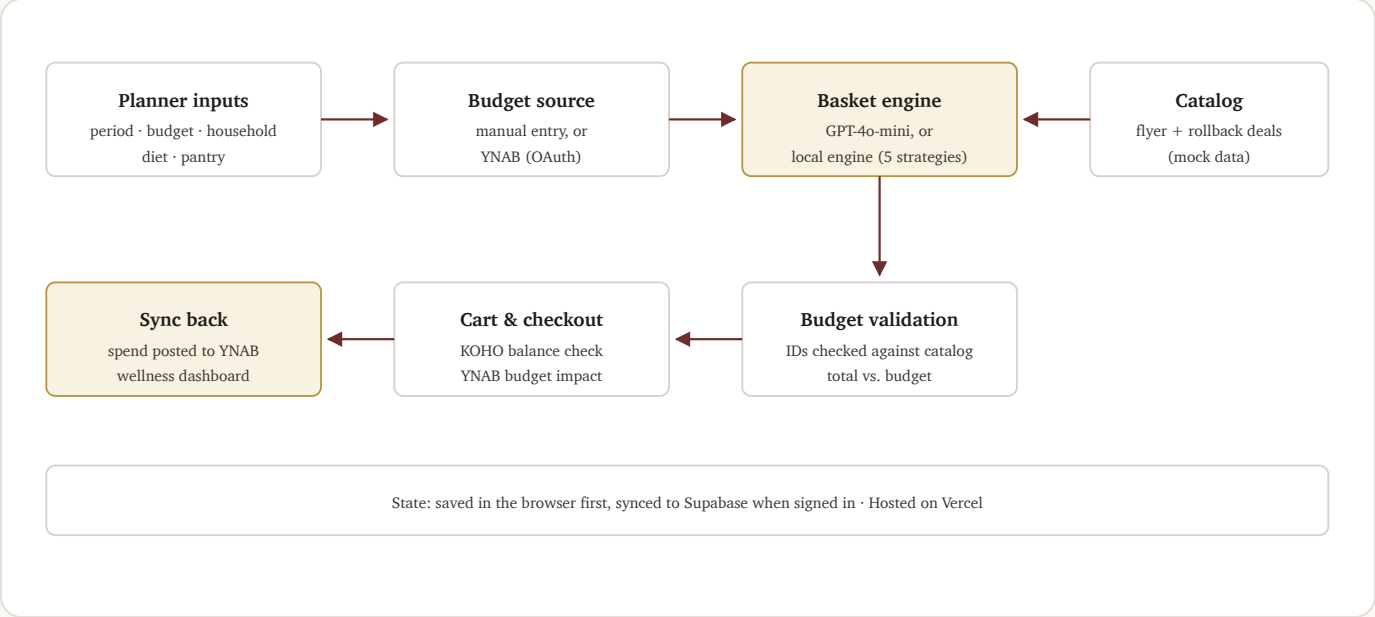
What to notice: the budget comes in before the basket is built, so the basket never starts over budget. Returning users skip onboarding because their preferences are recalled.

Failure states. If the budget API is unavailable, the shopper enters a budget manually. If flyer ingestion is unavailable, the planner uses a default optimized catalog.

SECTION 3

Architecture

How the MVP works end to end. The shopper's inputs and budget enter on the left, a budget-checked basket comes out on the right, and the spend flows back into YNAB after checkout.



What to notice: the AI can only pick items that exist in the catalog, and every pick is validated before it reaches the cart. If the AI call fails or no key is set, the local engine builds the basket instead.

Data sources

A **mock Walmart catalog** of grocery products with prices, rollback deals and ratings stands in for live data. Flyer ingestion is designed so the mock catalog can be swapped for **Walmart's product and flyer APIs** later without changing the screens.

Basket generation

OpenAI GPT-4o-mini receives the shopper's preferences and a compact copy of the catalog, and returns a basket as structured JSON. Its rules: pick only from the catalog using exact item IDs, stay within the total budget, respect dietary restrictions strictly, prefer items on sale, and cover the major food categories. Every returned ID is checked against the catalog. When no OpenAI key is set, or the call fails, the built-in **local engine** ranks the catalog against the shopper's preferences with five strategies: lowest cost, healthiest, bulk saver, family and brand.

Integrations

INTEGRATION	STATUS	NOTES
Supabase database	**Real**	Accounts and synced app state
YNAB OAuth and sync-back	**Real**	Pulls the grocery budget, posts the order back
Basket generation (local engine or OpenAI)	**Real**	Works without any key, using the local engine
Vercel deployment	**Real**	Hosts the live MVP
Walmart live product and flyer APIs	Conceptual	Mock catalog for now
KOHO / Plaid full integration	Conceptual	Demo mode in the MVP
Live payments and order processing	Conceptual	Simulated checkout

SECTION 4

Failure Modes & How They're Handled

Each entry starts from an assumption the product depends on and covers what breaks if it fails, how we'd notice and how we'd respond.

1 Users don't trust AI recommendations

Impact: High | Likelihood: Medium

Breaks: shoppers ignore the AI basket, so basket adoption stays low.

Detect: basket conversion declines.

Fix: hybrid controls with manual overrides, so shoppers can toggle, swap and switch strategies. Contain it by adding transparency and control.

2 Flyer data goes stale or wrong

Impact: High | Likelihood: Medium

Breaks: the planner recommends savings that don't exist at the shelf.

Detect: a rising price-mismatch rate.

Fix: SLA monitoring on flyer freshness (target: refreshed within 24 hours), with fallback to a default optimized catalog.

3 Budget sync is unreliable

Impact: Medium | Likelihood: Medium

Breaks: OAuth or YNAB API failures leave the planner without a budget.

Detect: the OAuth failure rate.

Fix: a manual budget fallback, so planning continues in manual mode without blocking the shopper.

4 Walmart APIs aren't accessible

Impact: High | Likelihood: High

Breaks: limited real-time integration with Walmart's product, flyer and cart systems.

Detect: integration blockers during the beta.

Fix: mock data behind a scalable architecture, plus a phased rollout that works within those limits.

5 AI recommendations are poor

Impact: High | Likelihood: Medium

Breaks: weak personalization; baskets don't match the household.

Detect: user trust declines.

Fix: feedback loops and confidence scoring, with recommendation retraining.

6 The AI service is unavailable

Breaks: the OpenAI call fails or no key is configured.

Detect: each basket is flagged as built by the AI or by the local engine.

Fix: automatic fallback to the local engine, so a basket is always generated.

SECTION 5

Metrics, Experiments & Trust

North Star

Weekly Repeat Grocery Planning Usage: the percentage of users who return every week to use the AI grocery planner. **Target: +25%** in repeat planner engagement within the first six months of the Walmart+ beta cohort. It measures habit, stickiness and Walmart+ retention potential. Groceries are bought every week, so a planner that works should become a weekly routine.

Success metrics

METRIC	TYPE	DEFINITION	TARGET	WHY THIS TARGET
Weekly repeat planning usage	North Star	% of users returning weekly to the AI planner	+25%	Ambitious but achievable for a tool that solves a weekly, recurring pain
Basket conversion rate	Business	% of AI baskets added to cart and moved toward checkout	+10–15%	Recommendation systems usually lift conversion step by step; a conservative, enterprise-realistic range
Walmart+ retention uplift	Business	Increase in Walmart+ retention among planner users	+8–12%	Retention moves gradually; this avoids overestimating the early impact
Budget compliance rate	User trust	% of baskets that stay within the stated budget	> 70%	A clear majority shows the core promise holds, while allowing for edits and price changes
Basket generation speed	Operational	Time to generate a basket	< 3 sec	Consumer AI tools need near-instant results; longer delays risk abandonment

Operational targets

BASKET GENERATION LATENCY	API UPTIME	CART ACCURACY	FLYER FRESHNESS SLA
< 3 seconds	> 99%	> 99%	< 24 hours

Experiments (Phase 1)

TEST	COMPARES	MEASURES
A	AI Planner vs. the standard Walmart grocery flow	Basket conversion · savings utilization · repeat usage
B	YNAB-synced budgets vs. manual budgets	Budget compliance · retention · user trust

Trust and privacy

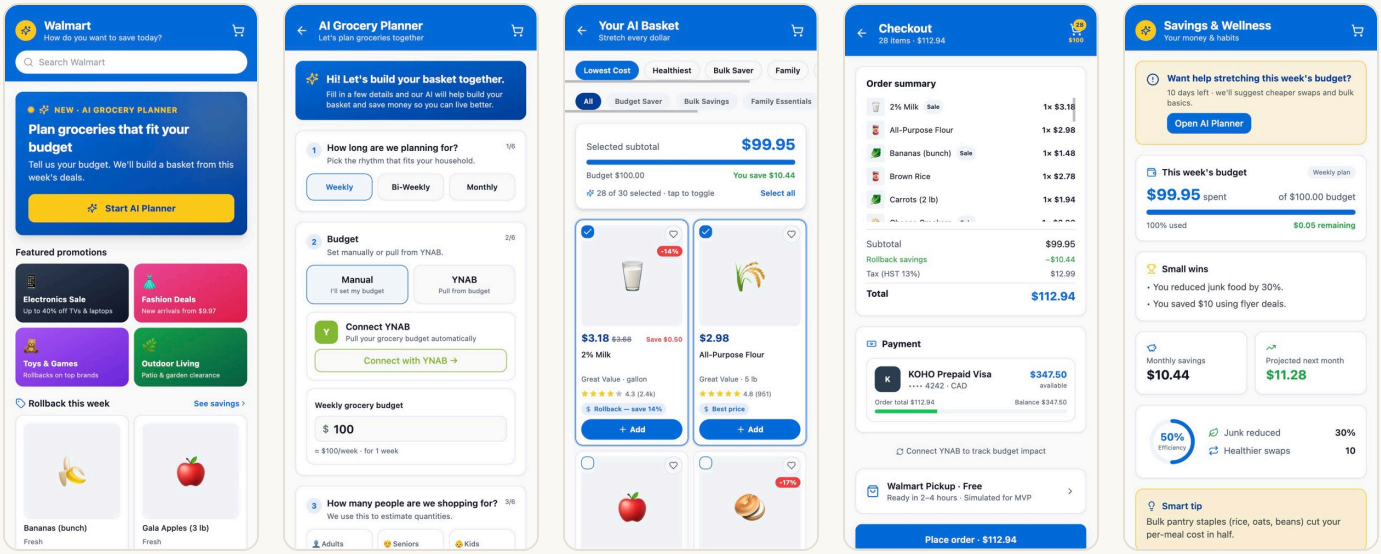
- **Transparent savings breakdown:** rollback savings are shown per item and in the order total.
- **Budget warnings:** the basket shows when the selection goes over budget, and by how much.
- **Privacy-first integrations:** YNAB and Plaid token exchanges run on the backend, so access tokens never reach the browser. In the MVP the OpenAI key is called from the browser to keep things simple; it moves to a server-side proxy before production.

SECTION 6

UX & Rollout

UX requirements

- **Walmart-native visual patterns**, so the planner feels like part of the Walmart app
- **Single-page onboarding** that covers period, budget, household, diet and pantry in one place
- **Persistent cart continuity** across screens and visits



HOME AI PLANNER AI BASKET CHECKOUT SAVINGS & WELLNESS

Screens from the live MVP running on mock data.

Rollout

PHASE	SCOPE
1	Walmart+ beta for budget-conscious households and families
2	Regional Walmart+ expansion, plus Smart Repeat
3	A full household grocery operating ecosystem

Milestones

MILESTONE	GOAL	STATUS
MVP prototype development	Core AI planner and Walmart workflow integration	**Completed**
PRD and strategic artifact buildout	PRD for stakeholders	**Current**
Beta integration phase	Walmart+ pilot deployment	Planned
Smart Repeat launch	Behavioral retention layer	Phase 2
Household grocery operating system	Full ecosystem expansion	Phase 3

Monetization

Primary: Walmart+ subscription retention, more repeat grocery transactions and larger recurring baskets. **Secondary:** fintech referral partnerships and premium AI planning tools. **Expected outcome:** higher customer lifetime value through retention and household reliance on the planner.

Product principles

Always prioritize: budget clarity · household simplicity · savings automation · user trust · configurable control · a Walmart-native experience · sustainable repeat behavior.

SECTION 7

AI Build Brief

The specification the basket planner is built against, taken from the MVP's planner.

Inputs

Planning period (weekly, bi-weekly or monthly) · budget (manual or from YNAB) · household (adults, seniors, kids) · dietary preferences · pantry items · the product catalog with prices, deals and ratings.

Rules the model must follow

1. Select only items that exist in the provided catalog, using exact item IDs.
2. Stay within the total budget for the planning period.
3. Respect dietary restrictions strictly.
4. Prefer items on sale when possible.
5. Cover the major food categories (produce, dairy, protein, pantry and others).
6. Return only valid JSON with a strategy, a short personalized explanation and the selected item IDs.

Requirements

Mapped from the RICE prioritization tiers in Section 2.

PRIORITY	REQUIREMENT
MUST	**AI Planner:** generate a personalized grocery basket from the shopper's inputs
MUST	**Budget Sync:** plan within the real budget, from YNAB or entered manually, and sync spend back
MUST	**Flyer Automation:** build baskets from this week's flyer and rollback deals
SHOULD	**Smart Repeat:** make next week's plan faster for returning households
SHOULD	**Wellness optimization:** encourage healthier choices within budget
COULD	**Behavioral insights** on spending and planning habits
COULD	**Pantry intelligence:** account for what the household already has

Guardrails

- Item IDs returned by the model are checked against the catalog; unknown IDs are dropped.
- If the model call fails or no key is set, the local engine builds the basket, so the shopper always gets one.

- The shopper always has the final say: items can be toggled, and strategies switched, before anything reaches the cart.

SECTION 8

Open Questions & Terms

Open questions

1. Will Walmart provide access to live product, flyer and cart APIs, and on what timeline? (Rated high impact and high likelihood in Section 4.)
2. Do the KPI targets hold with real users? They are planning estimates to be validated during the beta.
3. When should KOHO / Plaid move from demo mode to full integration?
4. When do AI calls move to a server-side proxy for production?

Explanation of terms

TERM	MEANING IN THIS DOCUMENT
Walmart+	Walmart's paid membership program. Its retention is the main business goal.
Flyer / rollback	Weekly promotional prices. "Rollback" is Walmart's name for a temporary price cut.
YNAB	"You Need A Budget", a budgeting app. Used here as a budget source and for spend sync-back.
KOHO	A Canadian prepaid Visa and money app. Its balance can be checked at checkout.
Plaid	A service that securely connects apps to bank accounts. Used to read the KOHO balance.
OAuth	A standard way to let one app access another without sharing a password. Used for YNAB sign-in.
North Star metric	The one metric that best captures the value the product delivers to users.
RICE	A prioritization method that scores features on Reach, Impact, Confidence and Effort.
A/B test	An experiment that compares two versions with similar user groups to see which performs better.
Budget compliance	Whether a generated basket stays within the stated budget.